

Occurrence of steroid hormones and antibiotics in shallow groundwater impacted by livestock waste control facilities.

Wastewater impoundments at concentrated animal feeding operations (CAFOs) represent a potential source of veterinary pharmaceuticals and steroid hormone contamination to shallow groundwater. This study investigates the occurrence of seventeen veterinary pharmaceuticals and thirteen steroid hormones and hormone metabolites in lagoons and adjacent groundwater at operating swine and beef cattle facilities. These sites were chosen because subsurface geology and previous monitoring of nitrate, ammonia and chloride levels in shallow ground water strongly indicated direct infiltration, and as such represent worst cases for ground water contamination by waste water. Pharmaceutical compounds detected in samples obtained from cattle facilities include sulfamerazine; sulfamethazine; erythromycin; monensin; tiamulin; and sulfathiazole. Lincomycin; ractopamine; sulfamethazine; sulfathiazole; erythromycin; tiamulin and sulfadimethoxine were detected in wastewater samples obtained from swine facilities. Steroid hormones were detected less frequently than veterinary pharmaceuticals in this study. Estrone, testosterone, 4-androstenedione, and androsterone were detected in wastewater impoundments at concentrations ranging from 30 to 3600ng/L, while only estrone and testosterone were detected in groundwater samples at concentrations up to 390ng/L. The co-occurrence of veterinary pharmaceutical and steroid hormone contamination in groundwater at these locations and the correlation between pharmaceutical occurrence in lagoon wastewater and hydraulically downgradient groundwater indicates that groundwater underlying some livestock wastewater impoundments is susceptible to contamination by veterinary pharmaceuticals and steroid hormones originating in wastewater lagoons. Copyright © 2010 Elsevier B.V. All rights reserved.